



# STROKE SYSTEM OF CARE

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## Program Design, Clinical Pathway, and Workflow

*Aligned with American Heart Association / American Stroke Association (AHA/ASA) Get With The Guidelines—Stroke and American Stroke Association International Comprehensive Stroke Center Certification*

Prepared for: [King Fahad Central Hospital]

Target Certification Level: [Comprehensive]

Document Owner: Stroke Program Medical Director & Stroke Coordinator

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## 1. Executive Summary

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This document defines the hospital's Stroke System of Care: the clinical pathway, interdepartmental workflow, governance structure, and performance measurement framework required to deliver timely, evidence-based stroke care and to achieve and sustain American Heart Association / American Stroke Association (AHA/ASA) and Joint Commission stroke center certification.

The program is designed around three principles that anchor every accreditation standard in this space: (1) rapid, protocol-driven recognition and treatment of acute stroke from symptom onset through discharge; (2) a defined, multidisciplinary chain of accountability across EMS, Emergency Department, Radiology, Neurology, Pharmacy, Nursing, and Rehabilitation; and (3) continuous data-driven quality improvement through the Get With The Guidelines–Stroke (GWTG-Stroke) registry.

- Standardize the prehospital-to-discharge stroke pathway across all points of entry.
- Meet or exceed AHA/ASA Target: Stroke time-based treatment benchmarks.
- Establish the governance, staffing, and data infrastructure required for the target certification level.
- Create a durable, auditable record for accreditation survey readiness.

## 2. Purpose and Scope

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This document applies to all patients presenting with signs or symptoms of acute stroke, transient ischemic attack (TIA), or suspected cerebral venous thrombosis (CVT/CVST), across all entry points into the organization, including Emergency Medical Services (EMS) transport, walk-in presentation, inter-facility transfer, and in-hospital (inpatient) stroke onset. It covers the full continuum: prehospital recognition, emergency evaluation, acute treatment (intravenous thrombolysis and endovascular thrombectomy where applicable), dedicated CVT/CVST diagnostic and anticoagulation pathways, inpatient management, secondary prevention, rehabilitation planning, and discharge transition.

The document is intended for use by the Stroke Program Medical Director, Stroke Program Coordinator, Emergency Department leadership, Neurology, Neurosurgery, Neurointervention, Radiology, Laboratory, Pharmacy, Nursing, Rehabilitation Services, EMS partner agencies, and the hospital Quality and Accreditation functions.

## 3. Accreditation Framework

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AHA/ASA stroke certification is administered in partnership with the Joint Commission (and, in some states, DNV, ACHC, or CIHQ) under four progressive certification levels. The pathway and workflow in Sections 5 and 6 are designed to satisfy the clinical elements common to all four levels, with the certification-specific elements noted where they diverge.

| Certification Level                      | Designed For                                                                                                 | Core Capability Beyond Prior Level                                                                                                       | Typical Governance Requirement                                                |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Acute Stroke Ready Hospital (ASRH)       | Hospitals or emergency centers with a dedicated stroke-focused program but limited inpatient stroke capacity | 24/7 stroke protocol activation, CT availability, IV thrombolysis capability, and a defined transfer agreement to a higher level of care | Designated stroke medical director and coordinator; written transfer protocol |
| Primary Stroke Center (PSC)              | Hospitals providing the critical elements for long-term improved outcomes with inpatient stroke unit care    | Dedicated stroke unit, 24/7 neurology availability, standardized order sets, and formal performance-measure reporting                    | Multidisciplinary stroke committee; program data abstractor                   |
| Thrombectomy-Capable Stroke Center (TSC) | Hospitals performing endovascular thrombectomy and providing post-procedural neurocritical care              | All PSC elements plus 24/7 neurointerventional and neurocritical care capability and dedicated post-procedure monitoring beds            | Neurointervention program oversight; case volume and outcome tracking         |
| Comprehensive Stroke Center (CSC)        | Hospitals providing the highest level of coordinated stroke care for the most complex patients               | All TSC elements plus 24/7 neurosurgery, advanced neuroimaging, dedicated neuro-ICU, and participation in stroke research                | Full multidisciplinary leadership team; research and outcomes infrastructure  |

Table 1. AHA/ASA and Joint Commission stroke certification levels (most to least advanced: CSC → TSC → PSC → ASRH).

Note: Certification requirements are periodically updated by the Joint Commission and AHA/ASA. Program leadership should validate all specific criteria, volume thresholds, and staffing requirements against the current edition of the Joint Commission Comprehensive Certification Manual for Disease-Specific Care before survey.

## 4. Program Governance and Structure

A certified stroke program requires a named, accountable leadership structure and a standing multidisciplinary committee that meets at a defined cadence to review performance data, case outcomes, and process gaps.

### 4.1 Core Leadership Roles

- Stroke Program Medical Director – physician champion accountable for clinical protocols, order sets, and outcome review.
- Stroke Program Coordinator – operational owner of workflow compliance, staff education, and registry data quality.
- Emergency Department Medical and Nursing Directors – accountable for door-to-treatment time performance.
- Neurology / Neurointervention / Neurosurgery Leadership – accountable for consultation availability and procedural readiness (level-dependent).
- Quality/Accreditation Liaison – accountable for survey readiness and regulatory alignment.

### 4.2 Stroke Committee

A multidisciplinary Stroke Committee — including EMS liaison, ED, Neurology, Radiology, Laboratory, Pharmacy, Nursing, Rehabilitation, and Quality — should convene at least quarterly to review GWTG-Stroke performance measures, case reviews, and improvement action plans, with minutes retained for survey evidence.

## 5. Clinical Pathway

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The pathway below represents the standardized, end-to-end route a stroke patient follows from symptom onset to discharge and secondary prevention. It is designed to be MECE across the patient journey — every arrival route and treatment decision maps to a single, unambiguous next step — and is the reference model for order sets, EMR build, and staff education.

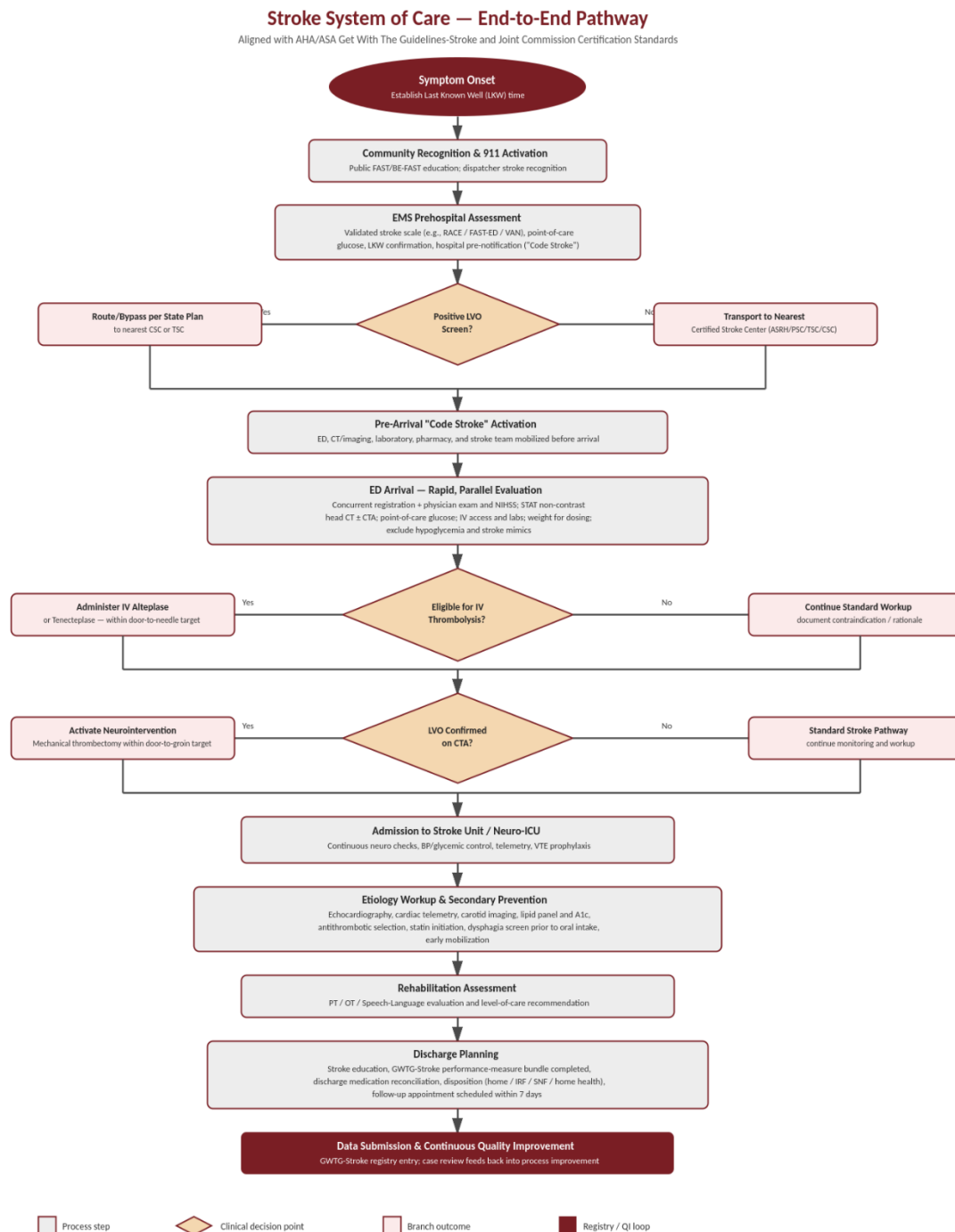


Figure 1. Stroke System of Care – end-to-end clinical pathway from symptom onset through registry submission.

## 5.1 Prehospital Phase

- Community education on FAST/BE-FAST symptom recognition and the importance of immediate 911 activation.

- EMS dispatch-level stroke recognition and priority response.
- On-scene validated stroke screening tool (e.g., RACE, FAST-ED, or VAN) to identify probable large-vessel occlusion (LVO).
- Last Known Well (LKW) time established and documented; point-of-care glucose obtained.
- Destination decision per state/regional EMS stroke triage and bypass protocol — direct transport to a Comprehensive or Thrombectomy-Capable center when LVO screen is positive and within transport parameters; otherwise transport to the nearest certified stroke center.
- Hospital pre-notification (“Code Stroke” alert) transmitted prior to arrival.

## 5.2 Emergency Department Phase

- Pre-arrival activation mobilizes the ED physician, stroke team, CT technologist, laboratory, and pharmacy before the patient arrives.
- Registration and clinical assessment occur in parallel, not sequentially.
- Standardized stroke order set initiated on arrival, including NIHSS, STAT non-contrast head CT (with CTA when LVO is suspected), point-of-care glucose, and weight-based dosing parameters.
- Eligibility for IV thrombolysis and/or endovascular thrombectomy determined per current AHA/ASA guideline criteria and documented, including rationale when treatment is withheld.
- When CVT/CVST is suspected, activate the appropriate Neurology/Stroke evaluation pathway and obtain venous neuroimaging (CTV or MRV) according to Radiology availability and patient factors; manage confirmed CVT/CVST under the dedicated institutional pathway rather than the routine arterial AIS reperfusion pathway.

## 5.3 Inpatient and Secondary Prevention Phase

- Admission to a dedicated stroke unit or neuro-ICU with continuous neurological monitoring, blood pressure and glycemic control, and telemetry.
- Dysphagia screening completed before any oral intake, medication, or food.
- Etiology workup (echocardiography, carotid imaging, extended cardiac monitoring), antithrombotic and statin therapy initiated per protocol, and VTE prophylaxis.
- Early PT/OT/Speech-Language evaluation to inform level-of-care and discharge disposition.

## 5.4 Discharge and Transition of Care

- Structured stroke education delivered to patient and caregiver before discharge.
- Discharge medication reconciliation and completion of the full GWTG-Stroke performance-measure bundle.
- Follow-up appointment scheduled, target within 7 days of discharge.
- Case entered into the GWTG-Stroke registry to close the quality-improvement loop.

# 6. Emergency Department Workflow and Time-Based Targets

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Time-based performance is the clinical and accreditation backbone of acute stroke care. The workflow below operationalizes the pathway into concurrent, rather than sequential, task assignments so that imaging, laboratory, pharmacy, and clinical assessment proceed simultaneously from the moment of arrival (“door”).

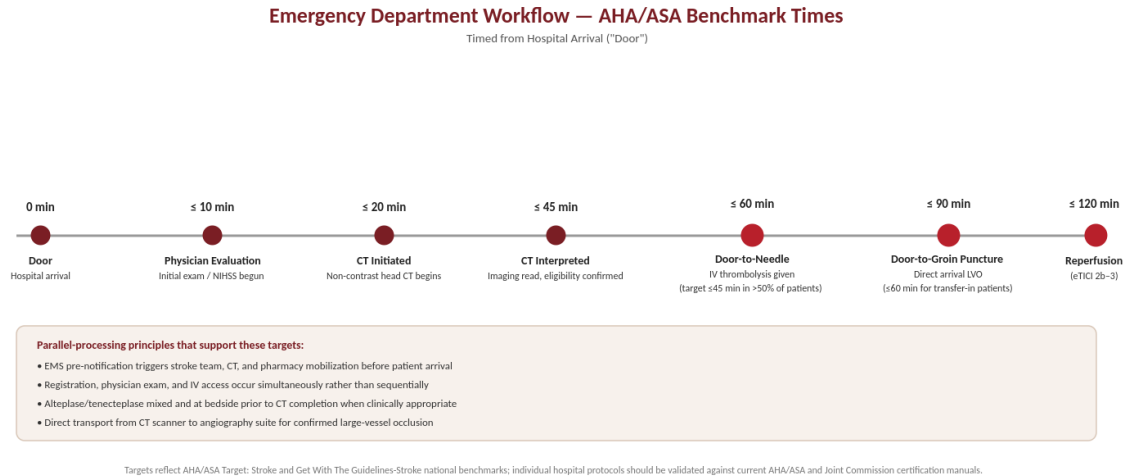


Figure 2. Emergency Department stroke workflow mapped against AHA/ASA Target: Stroke national benchmark times.

| Interval                                     | AHA/ASA Benchmark                               | Primary Accountable Function                |
|----------------------------------------------|-------------------------------------------------|---------------------------------------------|
| Door to Physician Evaluation                 | ≤ 10 minutes                                    | Emergency Department triage / physician     |
| Door to CT Initiation                        | ≤ 20 minutes                                    | Emergency Department / Radiology            |
| Door to CT Interpretation                    | ≤ 45 minutes                                    | Radiology                                   |
| Door to Needle (IV thrombolysis)             | ≤ 60 minutes (≤ 45 minutes in >50% of patients) | Emergency Department / Pharmacy / Neurology |
| Door to Groin Puncture (LVO, direct arrival) | ≤ 90 minutes                                    | Neurointervention team                      |
| Door to Groin Puncture (LVO, transfer-in)    | ≤ 60 minutes                                    | Neurointervention team                      |
| Door to Reperfusion (eTICI 2b–3)             | ≤ 120 minutes                                   | Neurointervention team                      |

Table 2. Core door-to-treatment benchmark times referenced by AHA/ASA Target: Stroke and Get With The Guidelines–Stroke.

Individual thresholds and eligibility criteria should be validated against the current AHA/ASA scientific statements and the hospital's target Joint Commission certification manual, as benchmark values are updated periodically.

## 7. Roles and Responsibilities

| Function             | Key Responsibilities in the Pathway                                                                     |
|----------------------|---------------------------------------------------------------------------------------------------------|
| EMS                  | Field stroke screen and LKW time; pre-notification; direct routing per regional bypass protocol         |
| Emergency Department | Code Stroke activation, rapid parallel assessment, order set initiation, IV thrombolysis administration |
| Radiology            | Immediate, prioritized imaging acquisition and interpretation for stroke alerts                         |



| Function                      | Key Responsibilities in the Pathway                                                      |
|-------------------------------|------------------------------------------------------------------------------------------|
| Neurology / Neurointervention | Rapid consultation, treatment-eligibility determination, thrombectomy when indicated     |
| Pharmacy                      | Rapid preparation and verification of thrombolytic dosing                                |
| Nursing (ED and Inpatient)    | Continuous neurological monitoring, dysphagia screening, care coordination               |
| Rehabilitation (PT/OT/Speech) | Early functional evaluation and discharge-disposition recommendation                     |
| Stroke Coordinator / Quality  | Workflow compliance monitoring, registry data abstraction, case review, survey readiness |

## 8. Performance Measurement and Continuous Quality Improvement

Performance is tracked through the GWTG-Stroke registry and reviewed at the Stroke Committee to drive an ongoing improvement cycle rather than a static, point-in-time compliance exercise.

1. Concurrent case identification and data abstraction for every stroke and TIA encounter.
2. Monthly dashboard review of door-to-treatment times and core performance measures (e.g., dysphagia screening, antithrombotic and statin therapy at discharge, VTE prophylaxis, stroke education, smoking-cessation counseling).
3. Case-level review of any patient who does not meet a time-based benchmark, with root-cause analysis.
4. Quarterly Stroke Committee review of aggregate trends and improvement action plans, with minutes retained as survey evidence.
5. Annual mock survey / readiness assessment against the current Joint Commission stroke certification manual.

## 9. Education and Training

- Annual stroke-specific competency validation for ED and inpatient nursing staff caring for stroke patients.
- Ongoing NIHSS certification/recertification for all clinical staff performing stroke assessments.
- Joint education sessions with EMS partner agencies on prehospital stroke scales and destination protocols.
- Community stroke-awareness education aligned with the certification level's outreach requirement.

## 10. Glossary of Acronyms

| Acronym | Definition                                               |
|---------|----------------------------------------------------------|
| AHA/ASA | American Heart Association / American Stroke Association |

| Acronym     | Definition                                                               |
|-------------|--------------------------------------------------------------------------|
| ASRH        | Acute Stroke Ready Hospital                                              |
| PSC         | Primary Stroke Center                                                    |
| TSC         | Thrombectomy-Capable Stroke Center                                       |
| CSC         | Comprehensive Stroke Center                                              |
| GWTG-Stroke | Get With The Guidelines–Stroke (AHA/ASA quality registry)                |
| LKW         | Last Known Well (time)                                                   |
| LVO         | Large Vessel Occlusion                                                   |
| NIHSS       | National Institutes of Health Stroke Scale                               |
| eTICI       | extended Thrombolysis In Cerebral Infarction (reperfusion grading scale) |
| VTE         | Venous Thromboembolism                                                   |
| EMS         | Emergency Medical Services                                               |